Approach to a Patient with Thyroid Disease

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Block: Endocrine/Urology/Reproduction

Thread: Clinical Medicine

Conflict of Interest: None

Learning Objectives

By exam time, you should be able to do the following:

1. Distinguish the clinical signs of hyper- and hypothyroidism and interpret thyroid hormone lab assays.
2. Identify etiologies of hyper- and hypothyroidism.
3. Generate the treatment options for hyper- and hypothyroidism.
4. Formulate the appropriate evaluation of a thyroid nodule and recognize the different types of thyroid cancer.

Study Questions

1. A 52-year-old postmenopausal cis woman presents with three months of fatigue to the point where she must take a nap most days. She has noticed hair thinning, and brittle nails. She experiences constipation and has gained approximately 10 pounds in the last 6 months. On exam, her vital signs are normal. Her patellar reflexes are diminished bilaterally. Which of the following patterns would you expect to find in this person?
   a. Elevated TSH, Elevated Free T4
   b. Elevated TSH, Suppressed Free T4
   c. Suppressed TSH, Elevated Free T4
   d. Suppressed TSH, Suppressed Free T4

2. A 23-year-old cis woman is seeing her primary care doctor for a new symptom of anxiety over the last 6 weeks. She has had palpitations, sweating, increased stool frequency, and shakiness. Her vitals show BP 169/99, HR 112, RR 12, Temp 99.1. On exam she is clammy. She has brisk reflexes and a fine tremor on outstretched hand. She has bulging eyes and inflammation around the eyes. She reports that this has been progressing for the past few months. Her primary care doctor obtains the following workup:
TSH: <0.05 mcIU/mL (ref: 0.45-4.5), Free T4 3.98 ng/dL (0.70-177), Total T3 ng/dL 301 (80-180). A urine pregnancy test is negative. An uptake and scan shows uniform uptake of radioiodine in the thyroid gland. Which of the following describes the mechanism of her hyperthyroidism?
   a. Stimulation of her TSH receptors by IgG
   b. Production of excess thyroid hormone by a thyroid nodule
   c. Inflammation of the thyroid gland leading to release of stored thyroid hormone
   d. Excess production of TSH by a pituitary adenoma

3. A 30 year old non-binary adult (assigned female at birth) with Graves disease comes to see you in clinic. They were diagnosed with Graves disease in childhood and were well controlled on methimazole until last year when they lost health insurance. They have recently reobtained health care coverage and are not currently taking any medications. On exam: BP 109/66, HR 68, RR 12, Temp 98.6
Recent labs show TSH: 0.08 mcIU/mL (ref: 0.45-4.5), Free T4 2.01 ng/dL (0.70-177), Total T3 ng/dL 222 (80-180), Positive TSI. They inform you that they plan to remove their IUD in the coming weeks so that they can attempt pregnancy. Which medication is most appropriate for this patient at this time?
   a. Methimazole
   b. Propylthiouracil
   c. Propranolol
   d. Prednisone

4. A 74-year-old cis man with a history of hypertension and diabetes comes to the emergency room after a fall with head injury. A CT Head in the Emergency Room rules out intracranial pathology but finds an incidental thyroid nodule. His TSH in the emergency room is 2.42 mcIU/mL (ref 0.45-4.5). He is discharged with follow up to his primary care doctor. Eight weeks later, you see him in primary care clinic. What is the next best step for workup of his thyroid nodule?
   a. TSH measurement
   b. Thyroid ultrasound
   c. Radioiodine Uptake and Scan
   d. MRI of the neck

5. A 14-year-old cis boy presents to your office for establishment of care. Multiple members of his family had thyroid cancer. He and his younger sister both had their thyroid removed prophylactically in early childhood to prevent thyroid cancer. He is very tall and thin with long limbs. Examination reveals numerous bumps on his lips and tongue. Which type of thyroid cancer is common for his medical condition?
   a. Papillary thyroid cancer
   b. Follicular thyroid cancer
   c. Medullary thyroid cancer
   d. Anaplastic thyroid cancer

Thyroid Basics

1. The thyroid sits at the base of the neck under the thyroid and cricoid cartilage along the midline.
2. The thyroid arises at base of the developing tongue. A thyroid diverticulum forms and migrates downward into the neck, connected to the tongue by a temporary, epithelialized tube called the thyroglossal duct. Normally, the duct involutes, however, in some individuals the duct fails to completely disappear, allowing for the formation of a thyroglossal duct cyst.
3. The functional cell of the thyroid is the follicular cell. Circulating iodide is oxidized and then conjugated to thyroglobulin in order to make thyroid hormone.
4. Triiodothyroxine (T3) is active thyroid hormone and Thyroxine (T4) is the major transport form. The thyroid releases T3:T4 in a 1:13 ratio. Circulating T4 is largely bound to proteins. Deiodinases in target tissue "activate" thyroid hormone by converting T4 to T3.
5. The most stable measurement of active thyroid hormone is "Free T4"

Thyroid Hormone Physiology

1. Thyroid hormone physiology functions like a typical endocrine feedback loop.
2. The hypothalamus secretes Thyroid Releasing Hormone (TRH) which causes the anterior pituitary to secrete Thyroid Stimulating Hormone (TSH). TSH, in turn, causes release of T3 and T4 from the Thyroid.
3. Small changes in serum Free T4 induce measurable changes in TSH through negative feedback.
4. In most cases, thyroid function is best evaluated using TSH. Abnormal TSH should prompt the collection of Free T4 and sometimes Total T3.
5. Measure free T4 if pituitary or hypothalamic disease is suspected.

Thyroid Hormone Function

1. Thyroid hormone works via a nuclear receptor, which means unbound thyroid hormone must translocate to the nucleus. It works by binding GNA and affecting gene expression and activation. Thus, thyroid hormone pathology often takes weeks/months/years to manifest clinically.
2. Think of the thyroid as the metabolic thermostat of the body. Almost all tissues are affected by thyroid hormone. Thyroid hormone globally increases catabolism – it increases heart and respiratory rate, energy expenditure, GI motility, cell turnover, etc.

Case 1

A 34-year-old cis woman presents with 3 months of weight loss, palpitations, insomnia, heat intolerance and increased stool frequency. She notes her menses have become lighter and more infrequent.

Her vital signs are remarkable for a mildly elevated systolic blood pressure with a pulse of 112 bpm.

Hyperthyroidism – Elevated Free T4, Elevated T3

1. Clinical Features
    - General – weight loss with retained appetite, sweating and heat intolerance.
    - Cardiovascular – palpitations (sinus tachycardia, atrial fibrillation).
    - Musculoskeletal – proximal myopathy, osteoporosis
    - GU – oligo or amenorrhea.
    - GI – hyperdefecation
    - Neurologic – nervousness, agitation, tremor, emotional lability.

2. Causes of Hyperthyroidism
    - Primary: Graves disease, Toxic Adenoma/Multinodular Goiter, Thyroiditis, Congenital
        - Suppressed TSH, Elevated Free T4 and Total T3
    - Central: TSH producing adenoma (very rare)
        - Elevated/Normal TSH, Elevated Free T4 and Total T3
    - Ectopic: Trophoblastic Disease/Multiple Gestation, Struma Ovarii
        - Suppressed TSH, Elevated Free T4 and Total T3
    - Exogenous: Excess Levothyroxine
        - Suppressed TSH, Elevated/Normal Free T4 and Total T3

Graves Disease

1. Graves disease – most common cause of hyperthyroidism. Autoimmune condition due to autoantibodies to the thyrotropin receptor. Thyroid stimulating antibodies (TSI) activate the receptor thereby stimulating thyroid hormone secretion from the entire thyroid. TSIs also promote goiter formation

2. Pathognomic features: Thyroid eye disease, pretibial myxedema, thyroid bruit, acropachy
3. TSI can be measured. It is an IgG and can cause the placenta and cause neonatal Grave's disease.
4. Cardinal features (triad) – hyperthyroidism, diffuse goiter +/- bruit, Ophthalmopathy +/- dermopathy.

Toxic Adenoma/Multinodular Goiter

1. Focal patches of hyperfunctioning follicular cells distended with colloid working independently of TSH. The excess thyroid production comes from the autonomous nodule.
2. Because TSH becomes suppressed, the surrounding normal thyroid does not get any stimulation and thus does not produce thyroid hormone.

Thyroiditis associated Hyperthyroidism

1. Thyroiditis: inflammation of the thyroid gland leading to leakage of premade thyroid hormone. Due to suppression of TSH, active thyroid hormone production from otherwise normal thyroid tissue stops.
2. Generally self-limited. Offer supportive care: Analgesics, Glucocorticoids (to control inflammation). Beta blockers can be used if tachycardia.
    a. Subacute thyroiditis: Usually associated with pain and tenderness + flu like signs.
    b. Silent thyroiditis: Similar pattern to subacute thyroiditis, but no pain
    c. Suppurative thyroiditis: An infection in the thyroid gland itself
    d. Postpartum thyroiditis: Can happen up to a year post-partum
    e. Drug-induced thyroiditis: Amiodarone, ICIs, IFNs, and Lithium

Primary Hyperthyroidism Workup

1. The pattern of primary hyperthyroidism often looks similar: Suppressed TSH and Elevated Free T4 and Total T3.
2. Graves antibody (TSI or TRAb) can be measured
3. An uptake and scan can tell you how thyroid hormone is being produced by the thyroid. Uniform thyroid hormone production suggests Graves. A focal or patchy image suggests an adenoma/multinodular goiter. Absent tracer uptake suggests thyroidits.

Primary Hyperthyroidism Treatment

1. Symptom Control: Beta blockers treat the symptoms of increased beta-adrenergic tone (tachycardia, tremors, palpitations, etc).
2. Thionamides block the production of thyroid hormone by blocking TPO. Can cause liver toxicity and agranulocytosis so hepatic function and CBC should be obtained before treatment. Dose based on Free T4 level. TSH takes months to recover.
    a. Methimazole: Usually first line. Lower risk of hepatic toxicity. Higher risk of teratogenicity in the first trimester
    b. Propylthiouracil: Preferred for people attempting pregnancy. Higher risk of fulminant hepatic necrosis.
3. Permanent Options:
    a. Thyroidectomy: Removal of half or entire thyroid. This can definitively treat Graves. Also can cure toxic adenoma or multinodular goiter. A total thyroidectomy requires lifelong hormone therapy.
    b. Radioactive Iodine Ablation: oral dose of radioactive iodine to destroy thyroid gland.

Misc Hyperthyroidism

1. Central hyperthyroidism due to a TSH adenoma is very rare. Generally the pattern is Normal TSH with an elevated Free T4 and Total T3. The normal TSH is "inappropriately normal" because an elevated Free T4 should suppress TSH production. Nonsuppressed TSH function suggests excess TSH is coming from the pituitary gland.
2. Struma Ovarii: Ectopic thyroid tissue in an ovarian teratoma
3. Exogenous: When thyroid hormone replacement is too high
4. Trophoblastic Disease/Twin Gestation: at high levels b-hCG can stimulate the TSH receptor.

Thyroid storm

1. Rare, life-threatening condition characterized by severe clinical manifestations of thyrotoxicosis
2. Typically in the setting of untreated hyperthyroidism
3. Triggered by acute stress (infection, surgery, trauma)
4. Altered mental status, fever, diarrhea, tachycardia, congestive heart failure
5. Treatment: beta blockers, thionamides, potassium iodide, glucocorticoids

Case 2

A 13 year old cis boy presents with symptoms of severe fatigue, constipation, cold intolerance, and brain fog for the past few years. His growth chart is shown. [see slide]

Hypothyroidism

Clinical Features

1. General: Weight gain, cold intolerance, lethargy
2. Cardiovascular: Bradycardia, diastolic hypertension
3. MSK: Arthralgia, poor growth
4. Neuro: Paresthesia, hyporeflexia
5. Reproductive: Menstrual disturbance
6. GI: Constipation, decreased appetite
7. Psych: Depression, hypersomnia, cognitive impairment
8. Skin: Dry skin, brittle nails, loss of outer third of eyebrow
    a. Myxedema: Generalized nonpitting edema due to GAG and water retention in the interstitial spaces.
9. Neck: Goiter, hoarseness

Causes of Hyperthyroidism

1. Primary hypothyroidism: Hashimoto's thyroiditis, iodine deficiency, infiltrative disease, congenital, thyroidectomy or radioactive iodine
    a. Elevated TSH, low/normal Free T4
2. Central Hypothyroidism: Pituitary issues (adenoma, craniopharyngioma, surgery, trauma), congenital
    a. Low TSH, low Free T4
3. Resistance: Genetic. Very rare. Not tested on USMLE

Hashimoto's Thyroiditis

1. Most common cause of hypothyroidism in the United States caused by autoimmune destruction of the thyroid gland.
2. Autoantibodies can be measured: TPO Ab, and Thyroglobulin Ab
3. Autoimmune Polyglandular Syndrome (APS1): defect in AIRE gene that causes multiple autoimmune endocrinopathies.

Treatment

1. In general, we give levothyroxine.
2. TSH is monitored to assess the adequacy of replacement in primary hypothyroidism.
3. Free T4 is monitored in patients with central hypothyroidism

Myxedema coma

- Medical emergency with a high mortality rate
- Severe, longstanding hypothyroidism
- Precipitated by an acute event in a poorly controlled hypothyroid patient
- Decreased mental status, hypothermia, hypotension, bradycardia, hyponatremia, hypoglycemia, and hypoventilation
- Treatment
    - Thyroid hormone, glucocorticoids (until the possibility of coexisting adrenal insufficiency has been excluded), supportive measures, management of coexisting problems

Case 4

A 76 year old cis man is admitted to the ICU for sepsis secondary to pneumonia and is on multiple vasopressors. The patient has tachycardia. Thyroid function is checked. TSH 0.2 mcIU/mL (ref: 0.4-4.5); Free T4: 0.6 ng/dL (0.7-1.77); Total T3 66 ng/dL (80-180)

Nonthyroidal Illness Syndrome (formerly euthyroid sick syndrome)

- Seen in critically ill patients, possible protective mechanism to prevent excessive tissue catabolism
- Low T3, low or low-normal T4, low, low-normal or normal TSH (similar pattern to patients with central hypothyroidism)

Case 4

A 50 year old trans woman is referred by their primary care doctor for evaluation of a right sided thyroid nodule. The nodule was seen incidentally on a CT scan of their neck obtained for evaluation of cervical disc disease. You palpate the patient's neck and you don't feel any thyroid nodules or lymphadenopathy.

What is your next step in management?

Thyroid Nodules

1. Prevalence is high ~1 in 6 people. They are very slow growing.
2. Causes of Thyroid Nodules
   - Benign – multinodular goiter, Hashimoto's thyroiditis, cysts, follicular adenomas, Hurthle cell adenomas
   - Malignant – Papillary thyroid cancer, follicular thyroid cancer, anaplastic thyroid cancer, medullary carcinoma (tumor of the parafollicular or C cell), primary thyroid lymphoma, metastatic carcinoma.
3. Evaluation: TSH (to assess for functional vs nonfunctional nodule), thyroid ultrasound
4. Discrimination of Malignancy –
   - Ultrasound
   - Suspicious Features: Composition (solid vs. cystic), Echogenicity (dark vs light), Shape (taller than wide), Margins (smooth vs. irregular), Echogenic foci (calcifications)
   - Ultrasound guided fine needle aspiration (FNA)
5. Types of thyroid cancer
   a. Differentiated Thyroid Cancer: Papillary thyroid cancer and follicular thyroid cancer. Both originate from thyroid follicular cells, they differ in appearance under microscope.
      i. Orphan Annie Cells and psammoma bodies are features of papillary thyroid cancer
   b. Medullary Thyroid Cancer: Originating from C-cells of the thyroid
      i. This is the type of cancer associated with multiple endocrine neoplasia type 2
6. Treatment of Thyroid cancer
   - It is a surgical disease.
   - +/- radioactive iodine.

Answers to Study Questions

1. **Answer B Elevated TSH, Suppressed Free T4**. This patient has classic symptoms of primary hypothyroidism: fatigue, weight gain, constipation, hair/nail changes, and delayed relaxation of reflexes. In primary hypothyroidism, the thyroid gland is not producing enough T4, so the pituitary responds by increasing TSH secretion. Elevated TSH with an elevated Free T4 (Answer A) is found in a TSH producing adenoma, which is associated with hyperthyroidism. Suppressed TSH with elevated Free 4 (Answer C) is found in hyperthyroidism conditions, such as Graves disease or toxic nodular disease. Suppressed TSH and Free T4 (Answer D) could indicate central hypothyroidism, however this patient has no other risk factors or evidence of pituitary/hypothalamic disease.

2. **Answer A: Stimulation of her TSH receptors by IgG**. She has overt hyperthyroidism with suppressed TSH and elevated T4/T3. The combination of young age, ophthalmopathy, and uniform uptake throughout the thyroid is classic for Graves disease. In Graves disease, TSI/TRAb antibodies bind and stimulate the TSH receptor, causing increased thyroid hormone synthesis and thyroid growth. Production of excess thyroid hormone by a thyroid nodule (Answer B) describes a toxic adenoma. Uptake would be focal rather than uniformly increased. Inflammation of the thyroid gland leading to release of stored thyroid hormone (Answer C) describes thyroiditis. Because preformed hormone is being released rather than newly synthesized, radioactive iodine uptake is typically low. Excess production of TSH by a pituitary adenoma (Answer D) describes a TSH-secreting pituitary adenoma. TSH would be elevated or inappropriately normal rather than suppressed.

3. **Answer B: Propylthiouracil**: Methimazole is generally preferred outside pregnancy, but this patient is actively planning pregnancy. PTU is preferred during the first trimester of pregnancy because methimazole exposure during early pregnancy is associated with a characteristic pattern of congenital abnormalities (methimazole embryopathy), including aplasia cutis and choanal/esophageal atresia. Therefore, for someone who is actively attempting pregnancy, PTU is the better choice while trying to conceive and during the first trimester. After the first trimester, patients are generally transitioned back to methimazole (Answer A) because PTU carries a risk of severe hepatotoxicity. Propranolol (Answer C) can help control adrenergic symptoms such as tremor and tachycardia, but it does not treat the underlying excessive thyroid hormone production. Her HR and blood

pressure are normal at this time. Glucocorticoids like prednisone (Answer D) can be used in certain severe thyroid conditions, such as thyroid storm, but are not appropriate definitive treatment for uncomplicated Graves hyperthyroidism.

4. **Answer B: Thyroid Ultrasound**. The next step in evaluating a thyroid nodule is thyroid ultrasound to characterize the nodule and determine whether FNA is indicated. Ultrasound assesses features such as composition, echogenicity, margins, calcifications, and shape. A TSH measurement (Answer A) does not need to be repeated. His TSH is 2.42, so there is no need to repeat it 8 weeks later. A radioiodine uptake and scan (Answer C) is generally indicated when the TSH is low/suppressed, because a hyperfunctioning nodule may explain the abnormal thyroid function. MRI (Answer D) is not part of the routine initial evaluation of a thyroid nodule.

5. **Answer C: Medullary Thyroid Cancer**. This patient has features of MEN2B, particularly the combination of familial thyroid cancer, prophylactic thyroidectomy in childhood, marfanoid habitus (tall, thin, long limbs) and mucosal neuromas: bumps on the lips and tongue. MEN2 is caused by activating mutations in the RET proto-oncogene and is strongly associated with medullary thyroid carcinoma (MTC). Papillary thyroid cancer (Answer A) is the most common thyroid cancer overall. Follicular and anaplastic thyroid cancers are not the characteristic thyroid cancer of MEN2.